

NORMAL DISTRIBUTION

FAST MCQ TRICKS — BY THUNDERSTUDY

STEP 1: The Magic Formula

No Z → No Answer! Always calculate Z-score first:

$$Z = (X - \mu) / \sigma$$

X = value, μ = mean, σ = standard deviation

TRICK 1: Mean is the CENTER

The curve is perfectly symmetric. Area on each side of mean = **0.5**.

Direct MCQ Answer: $P(X > \mu) = 0.5$ | $P(X < \mu) = 0.5$

TRICK 2: The 68–95–99 Rule (Saves Tables!)

Range	Percentage Area
$\mu \pm 1\sigma$	68%
$\mu \pm 2\sigma$	95%
$\mu \pm 3\sigma$	99.7%

Shortcut Memory: "1–2–3 → 68–95–99"

TRICK 3: Half-Area Ladder

- Mean to 1σ = **34%**
- 1σ to 2σ = **13.5%**
- 2σ to 3σ = **2.35%**

TRICK 4 & 5: Symmetry & Complements

Negative Z: $P(Z < -a) = P(Z > a)$. Table only has positive values, use symmetry!

Greater Than: $P(X > a) = 1 - P(X < a)$. Table gives left area only.

10-Second MCQ Method

1. Identify μ , σ , X and find Z .
2. Identify type: $<$ (Direct Table), $>$ ($1 - \text{Table}$), **Between** (Big - Small).
3. Eliminate impossible options (Prob < 0 or > 1).

EXAM SECRET (Topper Trick)

Most answers lie between **0.3 and 0.9**. If you get 0.02 or 0.99, recheck your steps quickly!

Match Z-score $\rightarrow$ Spot Table Value $\rightarrow$ Pick Winner

By Thunderstudy